

Port Scanning with NMap



Estimated time needed: 20 minutes

In this project you will install NMap in Kali Linux and run port scanning.

Learning Objectives

After completing this lab:

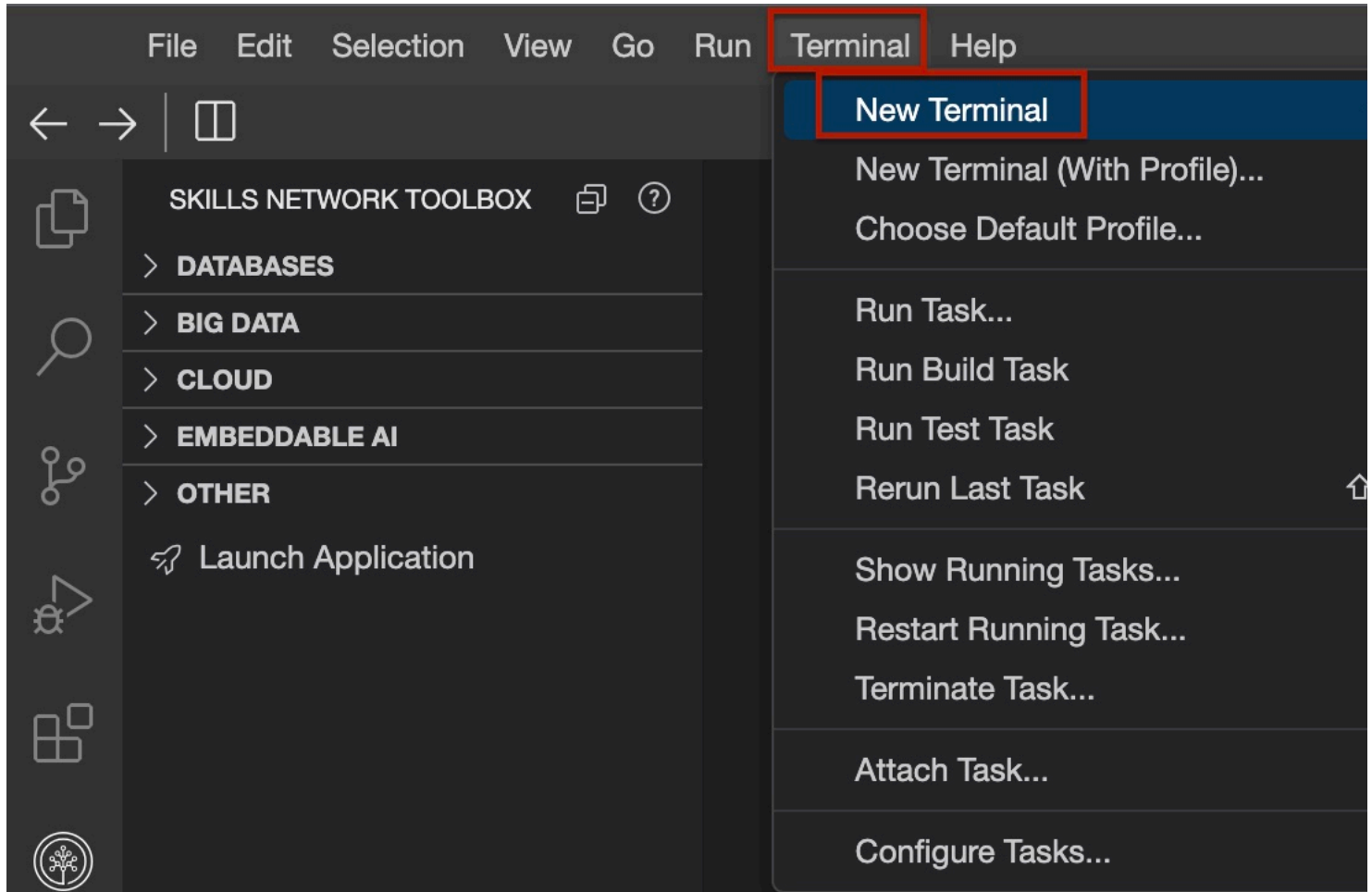
- Install NMap on Linux system
- Run port scan to check for vulnerabilities

Prerequisites

This lab requires docker which is preinstalled in Skills Network lab environment.

Set up Kali Linux environment

1. Open a new terminal.



2. In the new terminal that opens up, run the following command to obtain a docker image of Kali Linux with all the required commands preinstalled.

```
curl https://cf-courses-data.s3.us.cloud-object-storage.appdomain.cloud/9yTdNwJzgw_mSV1bDLT-1w/Dockerfile > Dockerfile
```

3. Build the Dockerfile in the current directory.

```
docker build . -t kalilinux
```

4. Run the Kali Linux from the docker image that was just built with an interactive shell.

```
docker run --tty --interactive kalilinux
```

This starts an interactive virtual Kali Linux environment. Any files you create in this environment will exist until you are in the virtual environment.

Install NMap

1. Run the following command to install NMap in Kali Linux.

```
apt-get update  
apt-get install -y nmap
```

This will take a few second to complete the installation.

2. Run the following command in the terminal to scan `scanme.nmap.org`.

```
nmap scanme.nmap.org
```

The output would be as seen below.

```
(root@0498317a5890)-[//]
# nmap scanme.nmap.org
Starting Nmap 7.94SVN ( https://nmap.org ) at 2023-12-15 00:33 UTC
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.0083s latency).
Other addresses for scanme.nmap.org (not scanned): 45.33.32.156 2600:3
Not shown: 999 filtered tcp ports (no-response)
PORT      STATE SERVICE
53/tcp    closed domain

Nmap done: 1 IP address (1 host up) scanned in 5.13 seconds
```

3. The default scan in nmap is SYN scan. It is a stealth scan to see if the target system has the ports specified in the command, open, closed or filtered. The SYN scan executes rapidly, capable of scanning thousands of ports within seconds on a high-speed network without hindrance from restrictive firewalls. To initiate a SYN scan using Nmap, the -sS option is utilized. -sS is commonly excluded when running Nmap as root or Administrator.

```
nmap -p22,113,139 scanme.nmap.org
```

4. Create a new user without root or admin privilege.

```
useradd John
```

5. Switch to the newly created user from root.

```
su John
```

6. Run the SYN scan for the ports 22,113 and 139 as John.

```
nmap -p22,113,139 scanme.nmap.org
```

But John is not a root user. To do SYN scan as a regular user, you need to provide -sS option to the command.

7. Run the following command to run SYN scan as a regular user.

```
nmap -p22,113,139 -sS scanme.nmap.org
```

As you can see, the regular user is disallowed to run the scan.

8. Exit the user prompt, to switch back to root user.

```
exit
```

8. To understand what nmap is doing at packet level, you can specify the `packet-trace` option. This gives you a detailed output.

```
nmap -d --packet-trace -p22,113,139 scanme.nmap.org
```

9. You can include higher debug levels to get more detailed output. Try the above command with `-d5`.

```
nmap -d5 --packet-trace -p22,113,139 scanme.nmap.org
```

Practice Exercise:

1. Try to nmap one of the public websites. eg., `google.com`.
2. Make a note of the open ports. Run nmap SYN scan on the open ports.
3. Do a packet-trace to the same ports that you have scanned above.

Conclusion:

Congratulations! In this lab, you have learnt how to perform nmap on a website, check for state of open ports and do a packet trace to specific ports.

Author(s)

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